

AP-E1: The Projective-Doublet Action Exact Quotient Theorems, Fixed-Charge Quantization, and the First-Principles Boundary of the Route-E Bridge

Route F research note

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Abstract

This note performs AP-E1 of the Another-Physics/Route-E program. The central result is a separation theorem. One nonzero charge-one complex doublet with a *local* common-phase redundancy and fixed norm has the exact quotient $S^3/U(1) = \mathbb{CP}^1$; eliminating the auxiliary connection gives the Fubini–Study sigma model. By contrast, a physical *global* common phase cannot be removed pointwise. At fixed global charge, a collective doublet reduces instead to a charged rotor on $T^*\mathbb{CP}^1$ in a unit Dirac-monopole background. Holomorphic sections $H^0(\mathbb{CP}^1, \mathcal{O}(k))$ arise only after a separate first-order symplectic reduction or a controlled lowest-Landau-level projection.

This distinction exposes a decisive Route-E blocker. The Hopf line has Chern number one, whereas $T_{\mathbb{CP}^1} \simeq \mathcal{O}(2)$. Neither $k = 2$, the holomorphic projection, nor the identification of its three states with three chiral families follows from the doublet quotient. Moreover, identifying the prequantum level with the existing macroscopic Q-ball benchmark gives $k = Q/\hbar = 10^6$ and $\dim H^0(\mathcal{O}(k)) = 1,000,001$, not three. A new separated-level construction is proposed: retain the macroscopic conserved charge and attach an independent level-two projective worldline sector. This is mathematically consistent and falsifiable, but its level-selection and UV origin remain explicit bridge axioms. A 30-check arithmetic and source regression audit verifies the quotient geometry, Chern number, Routh transform, spin-one algebra, signed rotor spectrum, Branch-B symplectic sign, Q-ball mismatch, and Derrick scaling without authorizing physics promotion.

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1 Scope, conventions, and final logical status

We use spacetime signature $(-+++)$, $\hbar$ explicitly in the quantum mechanics, and a single complex doublet

$$Z(x) = \begin{pmatrix} z_0(x) \\ z_1(x) \end{pmatrix} \in \mathbb{C}^2. \quad (1.1)$$

“One doublet” means two complex components. Two independent doublets form four complex components and, under one fixed-total-norm constraint and one common $U(1)$ quotient, generically give $\mathbb{CP}^3$, not $\mathbb{CP}^1$.

Four labels are used throughout.

Label	Meaning
proved	A consequence of the displayed action and hypotheses.
no-go	A stated set of hypotheses is incompatible with the claimed conclusion.
conditional	A consistent construction after an additional low-energy limit, projection, or sector choice.
bridge axiom	An identification or level-selection rule not implied by either the projective-doublet action or Route E.

Theorem 1.1 (AP-E1 separation theorem). *The following statements hold.*

1. *A fixed-norm complex doublet with a free local common-phase action has $\{Z : Z^\dagger Z = v^2\}/U(1) \simeq \mathbb{CP}^1$, and the canonical flat kinetic term induces the Fubini–Study metric.*

2. If the common phase is only global and the kinetic metric is nondegenerate, the phase is a local physical field. Quotienting one constant phase does not produce a pointwise $\mathbb{CP}^1$ field theory.
3. Fixing a global Noether charge and reducing a collective doublet gives a monopole-coupled $T^*\mathbb{CP}^1$ rotor. Its full Hilbert space contains infinitely many Landau levels.
4. $H^0(\mathbb{CP}^1, \mathcal{O}(k))$ is obtained from first-order Kähler reduction or from the lowest Landau level of that rotor. For $k = 2$ it is the spin-one triplet, but $k = 2$ and the projection are additional inputs.
5. The pure two-derivative $3 + 1$ -dimensional $\mathbb{CP}^1$ model has no finite-size static soliton. A four-derivative, flux, charge, wall, or gravity mechanism is required, and the full radial/gauge theory must still pass a stability test.

Promotion gate 1.2 (AP-E1 non-promotion). *AP-E1 may establish the projective geometry but may not promote the Route-E $\mathcal{O}(2)$ /three-family mechanism until a physical level-two source, a controlled holomorphic/LLL gap, a stable background, and a chiral fluctuation operator are all supplied. The numerical audit therefore keeps `physics_promotion_allowed` false.*

2 Minimal actions and the exact quotient

2.1 Auxiliary local action

Assume a global $SU(2)$ action on Z and a local common phase

$$Z \mapsto e^{i\alpha(x)}Z, \quad a_\mu \mapsto a_\mu + \partial_\mu\alpha, \quad D_\mu Z = (\nabla_\mu - ia_\mu)Z. \quad (2.1)$$

The exact fixed-norm auxiliary action is

$$S_{\text{aux}} = - \int d^4x \sqrt{-g} \left[(D_\mu Z)^\dagger D^\mu Z + \Lambda(x)(Z^\dagger Z - v^2) \right], \quad v > 0. \quad (2.2)$$

The multiplier equation fixes $Z^\dagger Z = v^2$, while the absence of a Maxwell term makes a_μ algebraic.

A renormalizable linear completion is

$$S_{\text{lin}} = \int d^4x \sqrt{-g} \left[-(D_\mu Z)^\dagger D^\mu Z - m^2 X - \frac{\lambda}{2} X^2 - \frac{1}{4e^2} f_{\mu\nu} f^{\mu\nu} \right], \quad X = Z^\dagger Z, \quad (2.3)$$

where $m^2 < 0$, $\lambda > 0$, and

$$v^2 = -\frac{m^2}{\lambda}. \quad (2.4)$$

Taking the radial stiffness large freezes X , while taking the vector kinetic scale heavy makes a_μ auxiliary at leading derivative order. For finite λ, e , the $\mathbb{CP}^1$ theory is an effective derivative expansion, not an exact equality of full theories. Gauged linear models give the standard action-level context for projective phases [1]; recent linear formulations of related Grassmannian models illustrate, but do not prove, a four-dimensional UV completion of the present model [2].

Remark 2.1 (What symmetry does and does not fix). *In a four-dimensional renormalizable scalar theory the canonical two-derivative term is fixed up to normalization after the field coordinates are chosen. Higher-dimensional EFT operators can deform the radial metric. After the quotient, however, every $SU(2)$ -invariant positive metric on $SU(2)/U(1) \simeq \mathbb{CP}^1$ is proportional to the round/Fubini–Study metric: the real two-dimensional tangent representation of the isotropy $U(1)$ is irreducible, so an invariant positive bilinear form is unique up to scale.*

2.2 Quotient theorem

Theorem 2.2 (Hopf quotient). *For $v > 0$, the vacuum manifold of (2.2) modulo the local common phase is*

$$\mathcal{M} = \{Z \in \mathbb{C}^2 : Z^\dagger Z = v^2\} / U(1) = S_v^3 / U(1) \simeq \mathbb{CP}^1 \simeq S^2. \quad (2.5)$$

Proof. The constraint surface is S_v^3 . Because $v \neq 0$, the common-phase action is free. Map each orbit to

$$P = \frac{ZZ^\dagger}{v^2}, \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1. \quad (2.6)$$

The projector is phase invariant. Conversely, a rank-one Hermitian projector selects a complex line in $\mathbb{C}^2$, and every unit vector spanning that line differs only by a phase. Rank-one complex lines are precisely $\mathbb{P}(\mathbb{C}^2) = \mathbb{CP}^1$. Writing

$$P = \frac{1}{2}(I + n^a \sigma_a), \quad n^a = \frac{Z^\dagger \sigma^a Z}{v^2}, \quad (2.7)$$

the Pauli completeness identity

$$(\sigma^a)_{ij} (\sigma^a)_{kl} = 2\delta_{il}\delta_{jk} - \delta_{ij}\delta_{kl} \quad (2.8)$$

gives $n^a n^a = 1$, identifying the quotient with S^2 . $\square$

This is also the elementary Marsden–Weinstein quotient $\mathbb{C}^2 // U(1)$ at a nonzero moment-map value [3]. It proves the quotient after the field content, charge, and nonzero level are stated; it does not derive those inputs.

3 Flat-doublet polar decomposition and induced geometry

3.1 One exact identity

On the patch $z_0 \neq 0$, set

$$N = 1 + |w|^2, \quad s_N(w) = \frac{(1, w)^T}{\sqrt{N}}, \quad Z = \rho e^{i\chi} s_N(w). \quad (3.1)$$

Define the real Hopf connection and Fubini–Study quadratic form by

$$\mathcal{A} = -i s_N^\dagger ds_N = \frac{i(wd\bar{w} - \bar{w}dw)}{2N}, \quad (3.2)$$

$$d\ell_{\text{FS}}^2 = ds_N^\dagger (I - s_N s_N^\dagger) ds_N = \frac{d\bar{w} dw}{N^2}. \quad (3.3)$$

Theorem 3.1 (Exact polar decomposition). *The flat Hermitian metric on $\mathbb{C}^2 \setminus \{0\}$ obeys*

$$dZ^\dagger dZ = (d\rho)^2 + \rho^2 [(d\chi + \mathcal{A})^2 + d\ell_{\text{FS}}^2]. \quad (3.4)$$

Proof. Since $s_N^\dagger s_N = 1$, differentiation gives $s_N^\dagger ds_N$ purely imaginary and $s_N^\dagger ds_N = i\mathcal{A}$. Decompose

$$ds_N = s_N (s_N^\dagger ds_N) + (I - s_N s_N^\dagger) ds_N. \quad (3.5)$$

The two terms are orthogonal, so

$$ds_N^\dagger ds_N = \mathcal{A}^2 + d\ell_{\text{FS}}^2. \quad (3.6)$$

Substitution in $dZ = e^{i\chi} [s_N d\rho + \rho (i s_N d\chi + ds_N)]$ gives (3.4); all radial cross terms cancel. $\square$

Thus the connection and the metric are not appended geometric decorations: they are respectively the vertical and horizontal parts of the flat doublet metric. In spherical coordinates

$$w = e^{i\phi} \tan \frac{\theta}{2}, \quad (3.7)$$

one obtains the convention-fixed formulas

$$\mathcal{A}_N = \frac{1 - \cos \theta}{2} d\phi, \quad (3.8)$$

$$\mathcal{F} = d\mathcal{A}_N = \frac{1}{2} \sin \theta d\theta \wedge d\phi, \quad (3.9)$$

$$d\ell_{\text{FS}}^2 = \frac{1}{4} (d\theta^2 + \sin^2 \theta d\phi^2) = \frac{1}{4} d\mathbf{n} \cdot d\mathbf{n}. \quad (3.10)$$

Our real line element therefore has sphere radius $1/2$, Gaussian curvature four, and scalar curvature eight. Other common complex-coordinate conventions place a factor of two in $ds^2 = 2g_{w\bar{w}}dw d\bar{w}$; all action, flux, and spectrum formulas here use (3.3).

3.2 Patch transition and Chern number

On $z_1 \neq 0$, let $\eta = 1/w$ and

$$s_S(\eta) = \frac{(\eta, 1)^T}{\sqrt{1 + |\eta|^2}}. \quad (3.11)$$

On the overlap, with $w = |w|e^{i\phi}$,

$$s_S = e^{-i\phi} s_N, \quad \mathcal{A}_S = \mathcal{A}_N - d\phi. \quad (3.12)$$

There is no global smooth Hopf section, but $\mathcal{F}$ is global and

$$\frac{1}{2\pi} \int_{\mathbb{CP}^1} \mathcal{F} = \frac{1}{2\pi} \int_0^{2\pi} d\phi \int_0^\pi \frac{1}{2} \sin \theta d\theta = 1. \quad (3.13)$$

This is the unit Hopf bundle. The sign of the line-bundle name depends on whether one associates charge $+1$ or -1 to the principal bundle; we fix the convention that the covariant derivative $D = d - ik\mathcal{A}$ has $c_1 = k$ and call it $\mathcal{O}(k)$.

3.3 Eliminating the auxiliary field

Varying (2.2) with respect to a_μ gives

$$a_\mu^\star = -\frac{i}{2Z^\dagger Z} \left[Z^\dagger \partial_\mu Z - (\partial_\mu Z^\dagger) Z \right]. \quad (3.14)$$

At fixed norm this becomes

$$a_\mu^\star = -\frac{i}{v^2} Z^\dagger \partial_\mu Z = \partial_\mu \chi + \mathcal{A}_\mu. \quad (3.15)$$

Substitution projects orthogonally to the gauge orbit:

$$\begin{aligned} (D_\mu Z)^\dagger D^\mu Z|_{a^\star} &= \partial_\mu Z^\dagger \left(I - \frac{ZZ^\dagger}{v^2} \right) \partial^\mu Z \\ &= v^2 \frac{\partial_\mu \bar{w} \partial^\mu w}{(1 + |w|^2)^2} = \frac{v^2}{4} \partial_\mu \mathbf{n} \cdot \partial^\mu \mathbf{n}. \end{aligned} \quad (3.16)$$

Hence

$$S_{\mathbb{CP}^1} = -v^2 \int d^4x \sqrt{-g} \frac{\partial_\mu \bar{w} \partial^\mu w}{(1 + |w|^2)^2}. \quad (3.17)$$

Its affine equation of motion is

$$\square w - \frac{2\bar{w}}{1 + |w|^2} (\partial_\mu w)(\partial^\mu w) = 0. \quad (3.18)$$

This classical projective construction is standard in the $\mathbb{CP}^{N-1}$ sigma-model literature [4].

4 How far the derivation is genuinely first-principles

There are two independent routes to the same metric, and their common boundary should be explicit.

4.1 Dynamical-field route

Assume:

1. a local, Lorentz-invariant, positive two-derivative theory;
2. exactly one complex order-parameter doublet with a global $SU(2)$ basis symmetry;
3. common phase is a local redundancy;
4. the radial mode is restricted to a nonzero level.

Then Theorem 2.2 and (3.16) force $\mathbb{CP}^1$ and the Fubini–Study kinetic term up to the stiffness v^2 . This is an action-level derivation, but the four assumptions are microscopic inputs.

4.2 Ray/fidelity route

Alternatively, suppose each point carries a normalized binary pure state $|u\rangle \in \mathbb{C}^2$, physical probabilities identify $|u\rangle \sim e^{i\alpha}|u\rangle$, and nearby states are compared by fidelity. Expanding

$$1 - |\langle u|u + du\rangle|^2 \quad (4.1)$$

to quadratic order gives

$$ds_{\text{fidelity}}^2 = \langle du|du\rangle - |\langle u|du\rangle|^2, \quad (4.2)$$

which is exactly (3.3). Pulling this information metric back to spacetime yields (3.17). This route explains why phase is a ray redundancy, but it still assumes a binary pure state and does not show that the ray is a fundamental spacetime field.

Corollary 4.1 (First-principles boundary). *Locality, Lorentz symmetry, positivity, and $SU(2)$ invariance determine the form of the low-energy action only after one complex doublet, ray equivalence, and a nonzero radial level are selected. They do not derive the number two, the scale v , or the microscopic origin of the order parameter.*

5 Global-only no-go theorem

Suppose instead that the common phase is a physical global symmetry. At fixed norm, (3.4) gives the exact local kinetic decomposition

$$\partial_\mu Z^\dagger \partial^\mu Z = v^2 \left[(\partial_\mu \chi + \mathcal{A}_\mu)^2 + \frac{\partial_\mu \bar{w} \partial^\mu w}{N^2} \right]. \quad (5.1)$$

Theorem 5.1 (Global $U(1)$ cannot give a pointwise projective target). *Assume the common phase is global, the kinetic metric is positive and nondegenerate, only one constant $U(1)$ orbit is quotiented, and no further constraint or gauge redundancy is introduced. Then the local target is not $\mathbb{CP}^1$.*

Proof. Equation (5.1) contains the independent local field $\chi(x)$. Quotienting $\chi \mapsto \chi + \alpha_0$ removes one constant zero mode, not arbitrary functions $\chi(x)$. Equivalently, the fixed-norm global theory has target S^3 , hence three local real coordinates, whereas $\mathbb{CP}^1$ has two. The Euler–Lagrange equation

$$\square\chi = -\nabla_\mu\mathcal{A}^\mu \quad (5.2)$$

shows that eliminating χ requires an inverse wave operator. Formally, it produces a nonlocal transverse term

$$\mathcal{A}_\mu \left(g^{\mu\nu} - \frac{\nabla^\mu \nabla^\nu}{\square} \right) \mathcal{A}_\nu, \quad (5.3)$$

not the pure local Fubini–Study action. $\square$

This is the first no-free-lunch statement: the same common phase cannot be simultaneously discarded as a local gauge redundancy and retained as the physical global charge that stabilizes an ordinary Q-ball.

6 Fixed global charge: exact collective Routh reduction

6.1 Doublet Q-ball ansatz

Consider a global $U(2)$ -invariant scalar theory in flat spacetime,

$$S_{\text{gl}} = \int d^4x \left[-\partial_\mu \Phi^\dagger \partial^\mu \Phi - U(\Phi^\dagger \Phi) \right]. \quad (6.1)$$

For the spatially coherent orientation ansatz

$$\Phi(\mathbf{x}, t) = f(\mathbf{x}) e^{i\chi(t)} s_N(w(t)), \quad (6.2)$$

define

$$\mathcal{I}[f] = 2 \int d^3x f^2, \quad E_0[f] = \int d^3x [(\nabla f)^2 + U(f^2)]. \quad (6.3)$$

Direct substitution, without using field equations, gives

$$L_{\text{coll}} = \frac{\mathcal{I}}{2} [(\dot{\chi} + \mathcal{A}_i \dot{q}^i)^2 + g_{ij}^{\text{FS}} \dot{q}^i \dot{q}^j] - E_0, \quad (6.4)$$

where q^i are real coordinates on $\mathbb{CP}^1$. The global charge is

$$Q = -i \int d^3x (\Phi^\dagger \dot{\Phi} - \dot{\Phi}^\dagger \Phi) = \mathcal{I}(\dot{\chi} + \mathcal{A}_i \dot{q}^i). \quad (6.5)$$

Consequently the fixed-profile energy, before minimizing the orientation velocity, is

$$E[f, q, \dot{q}] = E_0[f] + \frac{Q^2}{2\mathcal{I}[f]} + \frac{\mathcal{I}[f]}{2} g_{ij}^{\text{FS}} \dot{q}^i \dot{q}^j. \quad (6.6)$$

Thus the orientation-ground-state value (equivalently, the minimum over $\dot{q}$) is

$$E_Q^{\min}[f] = \min_{\dot{q}} E[f, q, \dot{q}] = E_0[f] + \frac{Q^2}{2\mathcal{I}[f]} = E_0[f] + \frac{Q^2}{4 \int f^2}. \quad (6.7)$$

This is compatible with the standard Q-ball variational principle [5, 6, 7]. It is exact after insertion of (6.2), but the ansatz is not yet proved to be a consistent truncation: spatially varying orientation, radial radiation, and nonspherical modes have been omitted. Fixed-charge EFT is independently testable in other theories, as recent lattice work confirms for the three-dimensional critical $O(2)$ model [8]; that evidence does not determine the coefficients of the present model.

6.2 Routh theorem

Theorem 6.1 (Fixed-charge projective rotor). *For*

$$L = \frac{\mathcal{I}}{2} [(\dot{\chi} + \mathcal{A}_i \dot{q}^i)^2 + g_{ij} \dot{q}^i \dot{q}^j] - V(q), \quad (6.8)$$

the Routhian at fixed Q and the reduced Hamiltonian are

$$R_Q = L - Q\dot{\chi} = \frac{\mathcal{I}}{2} g_{ij} \dot{q}^i \dot{q}^j + Q \mathcal{A}_i \dot{q}^i - V - \frac{Q^2}{2\mathcal{I}}, \quad (6.9)$$

$$H_Q = \frac{1}{2\mathcal{I}} g^{ij} (p_i - Q \mathcal{A}_i)(p_j - Q \mathcal{A}_j) + V + \frac{Q^2}{2\mathcal{I}}. \quad (6.10)$$

Proof. Solve (6.5) for

$$\dot{\chi} = \frac{Q}{\mathcal{I}} - \mathcal{A}_i \dot{q}^i \quad (6.11)$$

and substitute into $L - Q\dot{\chi}$, giving (6.9). Then

$$p_i = \frac{\partial R_Q}{\partial \dot{q}^i} = \mathcal{I} g_{ij} \dot{q}^j + Q \mathcal{A}_i, \quad (6.12)$$

and the Legendre transform yields (6.10). $\square$

The reduced classical phase space is the magnetic cotangent bundle $T^*\mathbb{CP}^1$, not merely $\mathbb{CP}^1$. For compact $\chi \sim \chi + 2\pi$, single-valued wavefunctions give

$$Q = \hbar k, \quad k \in \mathbb{Z}. \quad (6.13)$$

Under $s \mapsto e^{i\beta} s$, one has $\mathcal{A} \mapsto \mathcal{A} + d\beta$ and $\chi \mapsto \chi - \beta$. The reduced wavefunction is therefore a section with covariant derivative

$$D = d - ik\mathcal{A}, \quad D^2 = -ik\mathcal{F}, \quad \frac{i}{2\pi} \int D^2 = k. \quad (6.14)$$

7 Quantization, line bundles, and the Route-E obstruction

7.1 First-order symplectic reduction

Let $b = (b_0, b_1)^T \in \mathbb{C}^2$ have symplectic form

$$\Omega = i\hbar \sum_{A=0}^1 db_A \wedge d\bar{b}_A, \quad \mu = \hbar b^\dagger b. \quad (7.1)$$

For $k > 0$, at $\mu = k\hbar$, write

$$b = \sqrt{k} e^{i\chi} s_N(w). \quad (7.2)$$

The canonical one-form restricts, up to an exact term, to $k\hbar(d\chi + \mathcal{A})$, hence

$$\mu^{-1}(k\hbar)/U(1) = \mathbb{CP}^1 \quad (k > 0), \quad \Omega_{\text{red}} = k\hbar\mathcal{F}. \quad (7.3)$$

The zero level is exceptional:

$$\mu^{-1}(0) = \{0\}, \quad \mu^{-1}(0)/U(1) = \{\text{pt}\}; \quad (7.4)$$

it supplies the one-dimensional trivial representation, not a geometric $\mathbb{CP}^1$ orbit. Negative charge is obtained from the conjugate doublet (or the opposite symplectic orientation), and uses the anti-holomorphic polarization. Kähler quantization for $k > 0$, with $k = 0$ treated by the preceding trivial orbit, gives

$$\mathcal{H}_k = H^0(\mathbb{CP}^1, \mathcal{O}(k)) \simeq \text{Sym}^k(\mathbb{C}^2)^*, \quad \dim_{\mathbb{C}} \mathcal{H}_k = k + 1. \quad (7.5)$$

This is the clean finite-dimensional route. In Schwinger-boson language,

$$N = a_A^\dagger a_A = k, \quad J_i = \frac{\hbar}{2} a^\dagger \sigma_i a, \quad J^2 = \frac{\hbar^2}{4} N(N + 2). \quad (7.6)$$

For $k = 2$,

$$\mathcal{H}_2 = \text{Span}\{|2, 0\rangle, |1, 1\rangle, |0, 2\rangle\}, \quad j = 1, \quad \dim \mathcal{H}_2 = 3, \quad J^2 = 2\hbar^2. \quad (7.7)$$

The weights of $2J_3/\hbar$ are $(+2, 0, -2)$, exactly the Route-E weight pattern. This is a representation-theoretic match, not yet a physical family derivation.

7.2 Second-order rotor and the LLL gate

The second-order Hamiltonian (6.10) is different: it has all monopole harmonics [9, 10]. Since (3.10) is a sphere of radius $R = 1/2$, define $\kappa = |k|$. The magnitude of the monopole strength is $S = \kappa/2$, and

$$j = \frac{\kappa}{2} + n, \quad n = 0, 1, 2, \dots \quad (7.8)$$

The degeneracy and kinetic energy are

$$\begin{aligned} d_n &= 2j + 1 = \kappa + 2n + 1, \\ E_n^{\text{kin}} &= \frac{\hbar^2}{2\mathcal{I}R^2} \left[j(j+1) - \frac{\kappa^2}{4} \right] \\ &= \frac{2\hbar^2}{\mathcal{I}} \left[n(n + \kappa + 1) + \frac{\kappa}{2} \right]. \end{aligned} \quad (7.9) \quad (7.10)$$

Only the $n = 0$ lowest Landau level is isomorphic, for positive flux, to $H^0(\mathbb{CP}^1, \mathcal{O}(k))$. Negative flux uses the conjugate anti-holomorphic polarization and has the same spectrum with $\kappa = |k|$. The gap is

$$\Delta_{\text{LLL}} = E_1 - E_0 = \frac{2\hbar^2(\kappa + 2)}{\mathcal{I}}. \quad (7.11)$$

For $k = 2$, $d_0 = 3$ and $\Delta_{\text{LLL}} = 8\hbar^2/\mathcal{I}$. A finite-dimensional family carrier is therefore justified only when every relevant mixing, temperature, curvature, and portal scale is parametrically below this gap. Otherwise higher $d_n = 5, 7, \dots$ multiplets participate.

7.3 Why the projective line does not imply a level-two bundle

The Euler sequence on $\mathbb{CP}^1$ is

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1)^{\oplus 2} \longrightarrow T_{\mathbb{CP}^1} \longrightarrow 0, \quad (7.12)$$

so taking first Chern classes gives

$$T_{\mathbb{CP}^1} \simeq \mathcal{O}(2), \quad c_1(T_{\mathbb{CP}^1}) = 2. \quad (7.13)$$

But the minimal Hopf construction has $c_1 = 1$ by (3.13). Therefore

$$\boxed{\text{projective doublet} \implies \mathbb{CP}^1 \text{ but not physical prequantum bundle} = T_{\mathbb{CP}^1} = \mathcal{O}(2).} \quad (7.14)$$

One may impose that the physical fluctuation is tangent-valued, select a charge-two sector, or introduce an independent level-two Wess–Zumino term. Each is a new physical input. Coherent-state/Borel–Weil constructions explain the representation once the level is selected [11]; they do not select the level.

7.4 Macroscopic-charge contradiction

The existing bridge benchmark uses

$$Q = 10^6, \quad f_0^2 = 0.5 \text{ GeV}^2, \quad R = 64.9712654357 \text{ GeV}^{-1}. \quad (7.15)$$

In the step-profile collective ansatz,

$$\mathcal{I} = 2f_0^2 \frac{4\pi R^3}{3} = 1.1488215825 \times 10^6 \text{ GeV}^{-1}, \quad \frac{Q}{\mathcal{I}} = 0.870457184321 \text{ GeV}, \quad (7.16)$$

reproducing the stored angular frequency. If this same Q is used in (6.13), then

$$k = 10^6, \quad \dim H^0(\mathbb{CP}^1, \mathcal{O}(k)) = 1,000,001. \quad (7.17)$$

It cannot simultaneously be the $k = 2$ triplet. This is not a small correction: it separates two distinct physical sectors.

8 Static stability and the limits of topology

For a unit vector $\mathbf{n} : \mathbb{R}^3 \rightarrow S^2$, the two-derivative energy is

$$E_2 = \frac{v^2}{4} \int d^3x (\partial_i \mathbf{n})^2. \quad (8.1)$$

Use the size dilation $\mathbf{n}_\lambda(\mathbf{x}) = \mathbf{n}(\mathbf{x}/\lambda)$. In three spatial dimensions,

$$E_2(\lambda) = \lambda E_2, \quad E_0(\lambda) = \lambda^3 E_0. \quad (8.2)$$

Thus a positive $E_2 + E_0$ has

$$\left. \frac{dE}{d\lambda} \right|_{\lambda=1} = E_2 + 3E_0 > 0, \quad (8.3)$$

and shrinks; this is Derrick’s obstruction [12].

If finite gauge stiffness generates, at leading order,

$$\mathcal{F}_{\mu\nu} = \frac{1}{2} \mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n}), \quad (8.4)$$

$$\mathcal{L}_4 = -\frac{1}{4e^2} \mathcal{F}_{\mu\nu} \mathcal{F}^{\mu\nu} = -\frac{1}{16e^2} [\mathbf{n} \cdot (\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n})]^2, \quad (8.5)$$

then $E_4(\lambda) = \lambda^{-1} E_4$, and

$$E_2(\lambda) + E_4(\lambda) = \lambda E_2 + \lambda^{-1} E_4 \quad (8.6)$$

has a stationary minimum at

$$\lambda_\star = \sqrt{E_4/E_2}. \quad (8.7)$$

Finite-energy boundary conditions compactify space to S^3 , so

$$\pi_3(S^2) = \mathbb{Z}, \quad Q_H = \frac{1}{4\pi^2} \int_{S^3} \mathcal{A} \wedge \mathcal{F}. \quad (8.8)$$

The four-derivative model supports the familiar Hopf-energy mechanism and bounds [13, 14]. Two-component charged condensates provide an action-level route to similar low-energy structures [15]; recent preprint work studies local stability of large twisted configurations in two-flavor scalar QED [16]. These results motivate a numerical AP-E5 gate but do not constitute a global existence theorem for the present parameters.

Three caveats remain.

1. $\pi_0(\mathbb{CP}^1) = 0$, so the two poles are not topologically distinct vacua. An anisotropy such as $V_{\text{aniso}} = \Lambda_b^4(1 - n_3^2)$ creates two minima but explicitly breaks $SU(2)$.
2. If the radial mode can reach $Z = 0$, the projective coordinate is undefined and Hopf charge can unwind through a core.
3. Portal forces can create an upper stability charge; conserved $U(1)$ alone does not guarantee arbitrarily large stable Q-balls [17].

Exact Hopfion solutions in an inhomogeneous magnetic Hamiltonian further show that a length-setting deformation can evade scaling, but that condensed-matter Hamiltonian is not a Lorentz-invariant completion of AP-E1 [18].

9 Three completion branches

9.1 Branch A: physical charge two

Set the physical compact charge sector to $k = 2$. Equations (7.5)–(7.7) then give the triplet with no extra level field. This is the mathematically shortest route, but a two-quantum object is not in the macroscopic thin-wall regime of the current $Q = 10^6$ benchmark. It needs a genuine few-body/nonperturbative calculation rather than the classical Q-ball approximation. Non-Abelian Q-ball analyses show that internal charge sectors can be nontrivial [19], but they do not select this Route-E sector.

9.2 Branch B: separated macroscopic and projective charges

Retain a macroscopic Q-ball field Φ_Q with global charge Q , but introduce an independent worldline doublet $b(t)$ with local $U(1)_b$. A minimal first-order action is

$$S_{\text{sep}} = S_Q[\Phi_Q] + \int dt L_b + S_{\text{portal}}, \quad (9.1)$$

$$L_b = -\frac{i\hbar}{2} [b^\dagger D_t b - (D_t b)^\dagger b] + \hbar k c_t, \quad D_t b = (\partial_t - i c_t) b. \quad (9.2)$$

Under

$$b \mapsto e^{i\alpha(t)} b, \quad c_t \mapsto c_t + \dot{\alpha}, \quad (9.3)$$

the action changes only by the endpoint term $+\hbar k[\alpha(t_f) - \alpha(t_i)]$. The c_t equation imposes

$$b^\dagger b = k. \quad (9.4)$$

For closed histories, invariance of $e^{iS/\hbar}$ requires $k \in \mathbb{Z}$. Taking $k = 2$ gives $\mathcal{O}(2)$ independently of the macroscopic Q . Gauge-invariant portals may depend on

$$n_b^a = \frac{b^\dagger \sigma^a b}{b^\dagger b} \quad (9.5)$$

but must preserve the separate physical $U(1)_Q$ if the Q-ball is to remain stable.

This branch resolves the algebraic contradiction (7.17) and is the recommended AP-E1 continuation. It does *not* explain why the worldline level equals two or why this spin-one space carries chiral families. A pair-only microscopic constituent rule could make two the smallest nonzero allowed occupation, but that requires an explicit charge-two or discrete selection rule; it may not be asserted from $b \mapsto -b$ alone.

9.3 Branch C: tangent self-carriage

Postulate that the physical zero mode is a tangent deformation of the projective background. Then (7.13) forces $\mathcal{O}(2)$ and its three holomorphic sections. This is geometrically economical but moves the missing physics into the assertion “family mode equals tangent mode.” AP-E4 must derive that statement from a Dirac/fluctuation operator and exclude additional modes.

Branch	Exact gain	Remaining decisive test
A	One charge sector directly gives $H^0(\mathcal{O}(2))$.	Replace the macroscopic Q-ball approximation by a controlled two-quantum calculation.
B	Separates Q-ball stability charge from projective Chern level.	Derive the level-two worldline sector and show all portals remain below the LLL and charge-leakage gaps.
C	Euler sequence forces tangent bundle $\mathcal{O}(2)$.	Derive a chiral tangent-valued zero mode with no unwanted partners.

10 Reproducible numerical and algebraic audit

The script `route_f/code/verify_ap_e1_projective_doublet.py` writes `route_f/output/ap_e1_projective_doublet.json` and `route_f/output/ap_e1_projective_doublet.md`. Its

current result is

$$\boxed{30/30 \text{ checks pass, } \text{physics_promotion_allowed=false.}} \quad (10.1)$$

The checks include:

1. projector idempotence, Hermiticity, phase invariance, and the Hopf S^2 reconstruction;
2. exact vertical/horizontal metric decomposition, north/south chart agreement, and gauge covariance;
3. midpoint integration

$$\int \mathcal{F} = 6.283185468670607, \quad \frac{1}{2\pi} \int \mathcal{F} = 1.000000025702; \quad (10.2)$$

4. direct versus closed-form Routh and Legendre transforms;
5. the fixed-charge orientation-energy minimum, positive-level reduction domain, and charge-conjugation invariance of the signed spectrum;
6. $\dim \text{Sym}^k \mathbb{C}^2 = k + 1$ and its weight lists for $0 \leq k \leq 6$;
7. the $k = 2$ commutators and Casimir $J^2 = 2I_3$ to machine precision;
8. the Branch-B constraint and positive endpoint phase from its corrected first-order symplectic sign, plus a critical-source token/hash regression;
9. the first four monopole-rotor levels and $\Delta_{\text{LLL}} \mathcal{I} / \hbar^2 = 8$;
10. reconstruction of the stored Q-ball frequency and the 1,000,001-versus-three contradiction;
11. the independent E_2, E_4, E_0 Derrick scalings.

These are arithmetic and source regressions, not a proof checker for every TeX sentence. They test convention factors and known failure modes; they cannot establish a microscopic level-selection rule.

11 Claim ledger and ordered next steps

Claim	AP-E1 status	Reason
One local fixed-norm doublet gives $\mathbb{CP}^1$	proved	Free Hopf quotient and explicit projector bijection.
Canonical kinetic term gives FS geometry	proved	Exact flat-metric polar decomposition and auxiliary-field elimination.
Global-only phase gives pointwise $\mathbb{CP}^1$	refuted	Physical fiber mode remains; formal elimination is nonlocal.
Fixed Q gives a monopole rotor	proved conditionally	Exact in the coherent-orientation collective ansatz; full field-space truncation remains to be justified.
Rotor Hilbert space equals three states	refuted without LLL gate	Higher Landau levels give infinitely many states.
Doublet quotient forces $\mathcal{O}(2)$	refuted	Minimal Hopf Chern number is one; tangent Chern number is two.
$k = 2$ gives a triplet	proved	Borel–Weil/Schwinger algebra after the level is selected.
Q-ball benchmark supplies $k = 2$	refuted	Existing $Q = 10^6$ would give dimension 1,000,001.

Claim	AP-E1 status	Reason
Triplet equals three chiral families	bridge axiom	Requires the AP-E4 chiral operator, index, and spectral audit.
Pure $\mathbb{CP}^1$ bubble is stable	refuted	Derrick scaling; topology alone does not fix size.
Separated level-two sector is consistent	conditional	Action (9.1) is gauge-consistent, but its UV origin is open.

The recommended execution order is:

1. **AP-E2 regression:** retain the already proved discriminant–Killing/contact identity as an exact invariant check.
2. **AP-E3 level derivation:** implement Branch B and search for an anomaly, pair-selection, or compactification mechanism that fixes $k = 2$. Compute the actual Δ_{LLL} and reject the reduction if any portal or background scale is comparable to it.
3. **AP-E4 chirality:** construct the full fluctuation/Dirac operator on the candidate background; compute its index, kernel, partner modes, and spectral gap. A three-dimensional bosonic spin multiplet is not yet three chiral fermion families.
4. **AP-E5 stability:** solve the coupled radial, gauge, and orientation boundary-value problem at finite e, λ . Measure the full Hessian, charge-emission thresholds, and the path through $Z = 0$.
5. **AP-E6 onward:** only after AP-E3–5 pass may the projective sector be inserted into the Route-E messenger and phenomenology pipeline.

12 Conclusion

AP-E1 is closed at the geometry level and open at the physical-selection level. The strongest unconditional statement available from the displayed assumptions is

$$\boxed{\text{one local fixed-norm complex doublet} \implies \mathbb{CP}^1 \text{ with Fubini–Study dynamics.}} \quad (12.1)$$

The strongest no-go statement is

$$\boxed{\text{one global Q-ball phase} \not\Rightarrow \text{pointwise } \mathbb{CP}^1 \not\Rightarrow \mathcal{O}(2) \not\Rightarrow \text{three families.}} \quad (12.2)$$

The proposed separated-level action is the most coherent next hypothesis: it preserves the macroscopic stability charge while making the projective Chern level independently quantized. Its value $k = 2$, LLL isolation, chiral interpretation, and soliton stability are now sharply stated falsifiable gates rather than hidden assumptions.

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